

CIS 121 Introduction to Programming. Assignment 2 Problems

Develop an IPO Chart and C++ code the following problems. Upload the IPO and code files to Blackboard.

Save your files with the convention PS2P1, PS2P2 etc. PS1P1 is Problem set 1, program 1 etc.

1. Allow the user to enter two exam scores from the keyboard. The first exam is worth 60% of the total points and the second exam is worth 40%. Calculate the total score by multiplying each exam score input by the respective weighting then add the two results together. Display the total.

Input	Process	Output
• Score of Exam1 • Score of Exam2	• Declare a variable for the weight of exam1 and set as 60% (.60) • Declare a variable for the weight of exam2 and set as 40% (.40) • Assign the input for the score of Exam1 as a variable • Assign the input for the score of Exam2 as a variable • Calculate the product of score of Exam1 and weight of Exam1, assign result as a variable • Repeat calculations for Exam2 • Calculate the sum of weighted scores of Exam1 and Exam2, assign result as a variable • Display the variable to the user	• Total Score

2. Given the current stock price and quantity of stock, display the current value of the stock in your portfolio.

Input	Process	Output
• Stock Price • Quantity of Stocks	• Assign the input for the stock price as a variable • Assign the input for the quantity of Stocks as a variable • Calculate the product of the stock price and its quantity, assign result as variable • Display the product to user	• Value of the stock(s)

3. Enter the total for a meal. Compute a tip at 15%. Display total, tip and total with tip.

Input	Process	Output
• Total cost of the meal	• Declare a variable for gratuity% and set at 15% (.15) • Assign the input for the Total as a variable • Calculate the product of the Total and Gratuity/Tip, assign result as variable Tip • Calculate the sum of the Total and Tip, assign result as variable Total plus Tip • Display Total cost of the meal, Tip, Total plus Tip to user	• Total cost of the meal • Tip • Total plus Tip

4. The purchase price and current price of a stock is entered into your program.
Display the percentage increase or decrease of the stock.

Input	Process	Output
• Stock Purchase Price • Stock Current Price	• Assign the input for the Stock Purchase Price as a variable • Assign the input for the Stock Current Price as a variable • Calculate Price Change by finding the difference between Stock Current Price and Stock Purchase Price. • Divide Price Change by the Purchase Price, then multiply the result to 100 to find the Percentage Change • Display Percentage Change to user	• Percentage Change of the stock

5. You are setting up a business and need to compute the break even point. This indicates how many items you must sell at a given price to cover your overhead. Enter fixed costs, price per unit and cost per unit into your program. Compute the break even point by dividing fixed costs by the difference of price per unit and cost per unit.

Input	Process	Output
• Fixed Costs • Price per Unit • Cost per Unit	• Assign the input for the Fixed Costs as a variable • Assign the input for the Price per Unit as a variable • Assign the input for the Cost per Unit as a variable • Find the difference of Price/Unit and Cost/Unit, then divide from the Fixed Costs to find the Break-Even Point • Display the Break-Even Point	• Break-Even Point

Example Problems (do not have to do – solutions will be provided)

1. Get two numbers from the keyboard. Display the sum, product, difference and quotient of the two numbers.
2. Enter last name and credits taken. Tuition is \$250 per credit hour. Compute total tuition. Display last name and tuition.
3. Enter first name and number of steps walked in a day. For each step you burned .25 calories. Compute the number of calories burned. Display first name and calories burned.
4. Enter the name of the political party and number of votes for two political parties. Compute the percentage of votes each party achieved. Display the party and percentages of votes.